Worksheet # 1 Part B - Upload Solutions
Physics 2

ame: _____ Date: _____

Must show equations used and steps taken in determining the solution for each problem! No work No credit.

Upload the solutions to Questions 11 though 15

11) Vectors in component form.

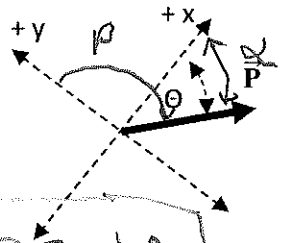
A) Given the vector $|\vec{P}| = 12$ N and the angle $\Theta = 40°$, determine the components of the vector.

$\vec{P} \cdot \hat{x} = P_x = |\vec{P}| \cos(\alpha) = 9.2N$

$\alpha = \Theta = 40°$

$\vec{P} \cdot \hat{y} = P_y = (\vec{P}) \cos(\beta) = -7.7N$

$\beta = 90° + \Theta = 130°$ | $\vec{P} = (9.2N)\hat{x} + (-7.7N)\hat{y}$



B) Force vector $\vec{C}$ is given in component form by $\vec{C} = (-500\text{ N})\hat{x} + (800\text{ N})\hat{y}$, determine the magnitude and direction (α & β) of vector $\vec{C}$?

$|\vec{C}| = \sqrt{C_x^2 + C_y^2} = 943.4N$

$\alpha = \cos^{-1}\left(\frac{C_x}{|\vec{C}|}\right) = 122°$ | $\beta = \cos^{-1}\left(\frac{C_y}{|\vec{C}|}\right) = 32°$

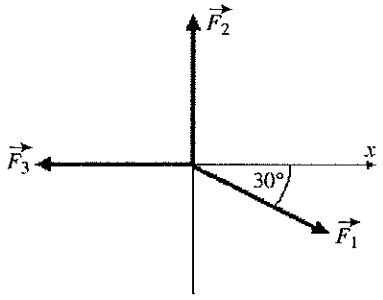
12) Adding vectors - As shown in the figure, three force vectors act on an object. The magnitudes of the forces as shown in the figure are $|\vec{F}_1| = 80$ N, $|\vec{F}_2| = 60$ N, and $|\vec{F}_3| = 40$ N. The net force acting on the object is given by $\Sigma\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3$. Determine the magnitude and direction of the net force acting on the object.

See Attached

$|\Sigma\vec{F}| = 35.5N$

$\alpha = 34°$

$\beta = 56°$



13) There are two spheres (A and B) with the same net charge (magnitude and type) and are separated by $r_1 = 0.3$ m, sphere B exerts a electrostatic force of 2 N on sphere A. The spheres are moved so that the separation is now $r_2 = 0.8$ m. Determine the net force that sphere A exerts on sphere B at the new separation of r_2. Assume the spheres are point charges.

need net charge

$|\vec{F}_{A\text{ on }B}| = |\vec{F}_{B\text{ on }A}|$

$\frac{kq_A q_B}{r_1^2} = \frac{kq_B q_A}{r_1^2} = |\vec{F}|$ $\Rightarrow$ $q = \sqrt{\frac{r_1^2 |\vec{F}|}{k}}$

$q = 4.47 \times 10^{-6}$ C

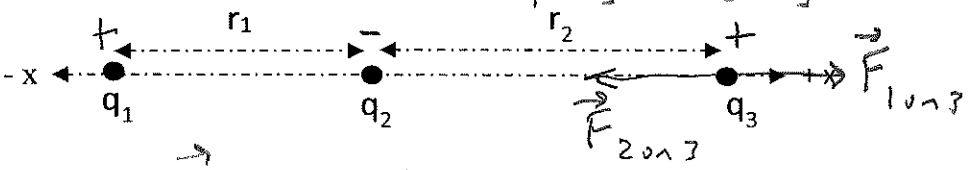
$|\vec{F}_2| = \frac{kq_A q_B}{r_2^2}$

$= 0.28$ N

14) Three point charges are placed on the *x*-axis. The separation, net charge and type of charge are given below. Determine the net electrostatic force exerted on **point charge q_3**.

$\Sigma \vec{F} = \vec{F}_{1\text{on}3} + \vec{F}_{2\text{on}3}$

$q_1 = 10$ μC and is positive
$q_2 = 15$ μC and is negative
$q_3 = 5$ μC and is positive
$r_1 = 0.1$ m and $r_2 = 0.2$ m



$|\vec{F}_{1\text{on}3}| = \frac{kq_1 q_3}{(r_1+r_2)^2} = 5$ N

$|\vec{F}_{2\text{on}3}| = \frac{kq_2 q_3}{r_2^2} = 16.9$ N

$\vec{F}_{1\text{on}3} = (-5\text{N})\hat{x}$

$\vec{F}_{2\text{on}3} = (16.9\text{N})\hat{y}$

$\Sigma \vec{F} = (\Sigma F_x)\hat{x} = (+11.9\text{N})\hat{x}$

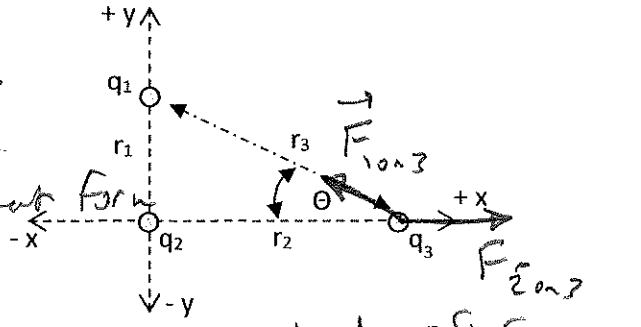
15) Three spheres with net charge are along the x and y axis, determine the magnitude and direction of the net force acting on **sphere 3 (q_3)?**

$\Sigma \vec{F} = \vec{F}_{1\text{on}3} + \vec{F}_{2\text{on}3}$

$q_1 = 10$ μC positive
$q_2 = 20$ μC negative
$q_3 = 30$ μC negative

$r_1 = 0.7$ m
$r_2 = 1.0$ m
$r_3 = 1.22$ m
$\Theta = 35°$

Need vectors in component form



magnitude of forces

See Attached

$|\Sigma \vec{F}| = 4.06$ N

$\alpha = 15°$

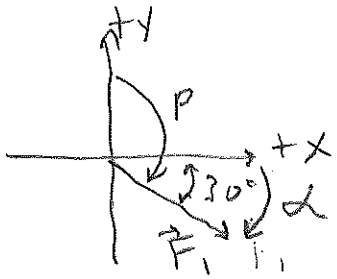
$\beta = 75°$

$|\vec{F}_{1\text{on}3}| = \frac{kq_1 q_3}{r_3^2} = 1.8$ N

$|\vec{F}_{2\text{on}3}| = \frac{kq_2 q_3}{r_2^2} = 5.4$ N

12) $\Sigma \vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 =$ net force

need vectors in component form



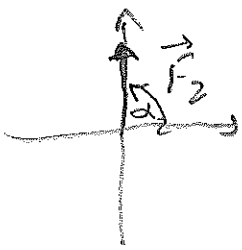
$\vec{F}_1 \cdot \hat{x} = F_{1x} = |\vec{F}_1| \cos(\alpha_1) = 69.3N$

$\alpha_1 = 30°$

$\vec{F}_1 \cdot \hat{y} = F_{1y} = |\vec{F}_1| \cos(\beta_1) = -40N$

$\beta_1 = 90° + 30°$

$\vec{F}_1 = (69.3N)\hat{x} + (-40N)\hat{y}$



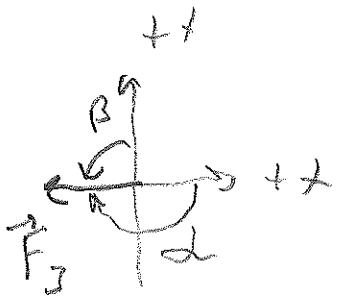
$\vec{F}_2 \cdot \hat{x} = F_{2x} = |\vec{F}_2| \cos(\alpha_2) = 0N$

$\alpha_2 = 90°$

$\vec{F}_2 \cdot \hat{y} = F_{2y} = |\vec{F}_2| \cos(\beta_2) = |\vec{F}_2| = 60N$

$\beta_2 = 0°$

$\vec{F}_2 = (0N)\hat{x} + (60N)\hat{y}$



$\vec{F}_3 \cdot \hat{x} = F_{3x} = |\vec{F}_3| \cos(\alpha_3) = -|\vec{F}_3| = -40N$

$\alpha_3 = 180°$

$\vec{F}_3 \cdot \hat{y} = F_{3y} = |\vec{F}_3| \cos(\beta_3) = 0N$

$\beta_3 = 90°$

$\vec{F}_3 = (-40N)\hat{x} + (0N)\hat{y}$

12) cont'

$$\Sigma \vec{F} = (\Sigma F_x)\hat{x} + (\Sigma F_y)\hat{y}$$

$$\Sigma F_x = \Sigma \vec{F} \cdot \hat{x} = 69.3N + 0N + (-40N) = 29.3N$$

$$\Sigma F_y = \Sigma \vec{F} \cdot \hat{y} = (-40N) + 60N + 0N = 20N$$

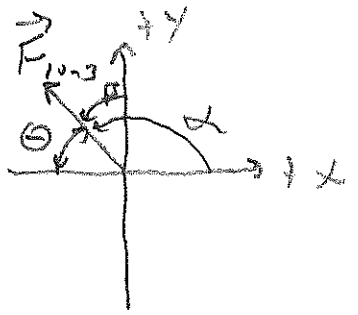
$$\Sigma \vec{F} = (29.3N)\hat{x} + (20N)\hat{x}$$

$$|\Sigma \vec{F}| = \sqrt{(29.3N)^2 + (20N)^2} = 35.48N$$

$$\alpha = \cos^{-1}\left(\frac{29.3N}{35.48N}\right) = 34^\circ$$

$$\beta = \cos^{-1}\left(\frac{20N}{35.48N}\right) = 56^\circ$$

15)



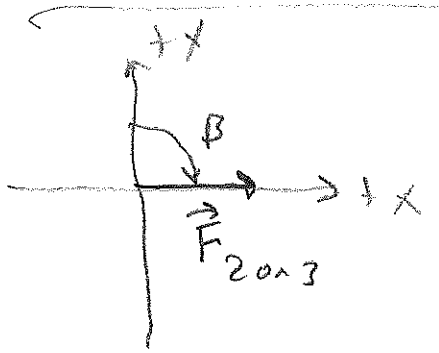
$\vec{F}_{1on3} \cdot \hat{x} = |\vec{F}_{1on3}| \cos(\alpha) = -1.47N$

$\alpha = 180° - \theta = 145°$

$\vec{F}_{1on3} \cdot \hat{y} = |\vec{F}_{1on3}| \cos(\beta) = 1.03N$

$\beta = 90° - \theta = 55°$

$\vec{F}_{1on3} = (-1.47N)\hat{x} + (1.03N)\hat{y}$



$\vec{F}_{2on3} \cdot \hat{x} = |\vec{F}_{2on3}| \cos(\alpha) = |\vec{F}_{2on3}|$

$= 5.4N$

$\alpha = 0°$

$\vec{F}_{2on3} \cdot \hat{y} = |\vec{F}_{2on3}| \cos(\beta) = 0N$

$\beta = 90°$

$\vec{F}_{2on3} = (5.4N)\hat{x} + (0N)\hat{y}$

$\Sigma \vec{F} = (\Sigma F_x)\hat{x} + (\Sigma F_y)\hat{y}$

$\Sigma F_x = (-1.47N) + (5.4N) = 3.93N$

$\Sigma F_y = 1.03N + 0N = 1.03N$

$\Sigma \vec{F} = (3.93N)\hat{x} + (1.03N)\hat{y}$

$|\Sigma \vec{F}| = \sqrt{\Sigma F_x^2 + \Sigma F_y^2} = 4.06N$

$\alpha = \cos^{-1}\left(\frac{\Sigma F_x}{|\Sigma F|}\right) = 15°$ $\beta = \cos^{-1}\left(\frac{\Sigma F_y}{|\Sigma F|}\right) = 75°$